

Machine Learning

LAB EXERCISE 5

This lab exercise must be submitted by May 30nd, 2025 at 10:00 am.
Late submissions will not be accepted and will be marked as zero.

This lab exercise is mandatory and could be assessed.
It is worth 1% of your total final grade for this course.

Submission Instructions

Submit your files through EClass. The files you submit cannot be read by any other students. You can replace your submission as many times as you like by resubmitting it, although only the last version sent is kept.

If you have last-minute problems with your EClass submission, email your assignment as an attachment to alberto.paccanaro@fgv.br with the subject "URGENT – LAB 5 SUBMISSION". In the body of the message, explain the reason for not submitting it through EClass.

All the work you submit should be solely your own work. Coursework submissions will be checked for this.
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In this lab, you will implement the Random Forest algorithm for both regression and classification. Depending on the problem, your trees will use nominal or quantitative features.

DATASETS DESCRIPTION

Energy Efficiency Dataset: This dataset (in file ENB20212_data.csv) is constituted by 768 points representing different buildings shapes and is associated to the prediction of the energy load required to cool/heat the building. The first 8 columns represent different features of the building:

- X1 – Relative compactness
- X2 – Surface Area
- X3 – Wall area
- X4 – Roof area
- X5 – Overall Height
- X6 – Orientation
- X7 – Glazing Area
- X8 – Glazing Area Distribution

This dataset is associated with 2 regression tasks: predicting Y1 (heating load) in the 9th column; and predicting price Y2 (cooling load) in the 10th column.

Car dataset: this dataset is constituted by 1728 points in 6 dimensions. Each point refers to a car, which is described by 6 attributes, all of them (first 6 columns). Each car belongs to one of possible 4 classes (last column).

The description of the attributes is as follows

- buying: buying price
- maint : price of the maintenance
- doors : number of doors
- persons: capacity in terms of persons to carry
- lug_boot: the size of luggage boot
- safety : estimated safety of the car

The values of the attributes are:

- Buying: v-high, high, med, low
- Maint: v-high, high, med, low
- Doors: 2, 3, 4, 5-more
- Persons: 2, 4, more
- lug_boot: small, med, big
- safety: low, med, high

YOUR TASKS

Build two random forests for the energy efficiency dataset (one for the Y1 attribute and the second for the Y2 attribute).

Build a random forest for the car dataset (here you should predict the class of the car using the categorical features)

You will need to keep an eye on a few important details:

- The decision trees should all be binary, even when your data contains nominal attributes that can assume more than 2 values (remember that for any non-binary tree there is always a binary one which is equivalent).
- Properly estimate your hyperparameters using using a validation set.

Have fun ! 😊